

TCP Congestion Control

Data Network — Transport Layer

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1 (Overview)

TCP Congestion Control (congestion), -- (end-to-end)
 . IP , TCP (loss event) (implicit) .

Note

Flow Control vs. Congestion Control

Flow Control	Congestion Control
<i>rwnd</i> (Receive Window) Sender Receiver	(Timeout / 3 Dup ACK)

1.1 “Router” ?

“router”, “router” . Switch , router ?
 ()

Switch L2() MAC , Router L3() IP . TCP Transport Layer ,
 TCP “ ” IP , **router** . TCP router .

Switch . TCP “IP ” router .

IP layer ()

IP best-effort , TCP (ECN). TCP router , / **RTT**
 . “end-end congestion control (no network assistance)” .

Tip

. Router **L3 IP** , IP TCP .

1.2

$$\text{LastByteSent} - \text{LastByteAcked} \leq \min(\underbrace{cwnd}_{\text{congestion window}}, \underbrace{rwnd}_{\text{flow control window}})$$

- **cwnd (Congestion Window):** unACKed . .

- **ssthresh (Slow Start Threshold):** Slow Start Congestion Avoidance .
- :

$$\text{rate} \approx \frac{cwnd}{RTT} \text{ [bytes/sec]}$$

2 cwnd

2.1 cwnd ?

Congwin = **Congestion Window (cwnd)** . “ ” , .

2.2 cwnd ?

TCP sender (invariant) :

$$\text{LastByteSent} - \text{LastByteAcked} \leq cwnd$$

, “ (in-flight)” (unACKed) cwnd . cwnd ACK .

$$\approx \frac{cwnd}{RTT} \text{ [bytes/sec]}$$

2.3 cwnd — cwnd ?

4 :

send_base	Already ACKed ()
send_base ~ nextseqnum	Sent, not yet ACKed (ACK)
nextseqnum ~ cwnd	Usable, not yet sent ()
cwnd	Not usable ()

cwnd “ (upper bound)” , . (usable, not yet sent) Application , .

i Note

Application cwnd . “cwnd” .

3 (Costs of Congestion)

TCP , .

3.1 Scenario 1: (,)

: (A, B) R . , .
 (λ_{in}) $R/2$ (delay) .

$$\lim_{\lambda_{in} \rightarrow R/2} \text{delay} = \infty$$

delay ?

. μ , λ :

$$E[\text{delay}] \propto \frac{1}{\mu - \lambda}$$

R $R/2$. sender $\lambda_{in} \rightarrow R/2$ R .

💡 Tip

$$\begin{array}{c} \text{ : } \\ \frac{\lambda_{in}}{\ll R/2} \text{ delay} \\ \rightarrow R/2 \quad () \rightarrow \infty \end{array}$$

: throughput $R/2$, delay . “ ” .

3.2 Scenario 2: ()

: , .

:

(Perfect Knowledge): sender $\rightarrow$, goodput = offered load.

(Known Loss): goodput $R/2$ overhead .

$$\lambda_{out} < \lambda'_{in} \quad (\lambda'_{in} = \text{original} + \text{retransmitted})$$

: (Premature Timeout): sender () $\rightarrow$ $\rightarrow$ goodput $R/4$.

Warning

(positive feedback)
(Scenario 2): 1. goodput 2. goodput ($\lambda_{out} < \lambda_{in}$)

3.3 Scenario 3: ()

: 4 , 2-hop , .

Host A $\rightarrow$ [R1] $\rightarrow$ [R2] $\rightarrow$ Host C

$\uparrow$

Host D $\rightarrow$ Host B

A $\rightarrow$ C R1, R2 . D $\rightarrow$ B R2 .

- R2 A $\rightarrow$ C D $\rightarrow$ B
- D $\rightarrow$ B R2 A $\rightarrow$ C
- R1 A $\rightarrow$ C , R2

Warning

Scenario 3 :
“upstream , downstream .”
 λ'_{in} (A $\rightarrow$ C) D $\rightarrow$ B goodput 0 .

:

Scenario 1	Scenario 2	Scenario 3
delay $\rightarrow \infty$	goodput	
goodput $R/2$ ()	$< R/2$ ()	$\rightarrow 0$ ()

4 TCP

TCP (state) .

4.1 Slow Start (SS)

$cwnd$ (exponential) .

:

- : $cwnd = 1 \text{ MSS}$
- **ACK** : $cwnd += 1 \text{ MSS}$
- : $RTT \quad cwnd \cdot 2$

$$cwnd(t + 1 \text{ RTT}) = 2 \cdot cwnd(t)$$

(exponential) ?

“ 1 ” . $RTT = 1$ **ACK** 1 1 .

$cwnd = N \text{ MSS}$ $RTT = N$, **ACK** N . **ACK** $cwnd += 1 \text{ MSS}$:

$$cwnd_{RTT} = cwnd + \underbrace{N}_{ACK} \times 1 \text{ MSS} = cwnd + cwnd = 2 \times cwnd$$

$RTT = 2$. $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow \dots$ — .

i Note

“ $RTT = 1$ ” , “**ACK** 1” $\rightarrow$ “ $RTT = cwnd$ ” .

:

$cwnd \geq ssthresh$
Timeout ()

Congestion Avoidance
 $ssthresh = cwnd/2$, $cwnd = 1 \text{ MSS}$, SS

3 Duplicate ACKs

$ssthresh = cwnd/2$, $cwnd = ssthresh + 3$, Fast Recovery

```

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

def simulate_tcp(initial_ssthresh=16, max_rounds=25, loss_at=None):
    cwnd = 1
    ssthresh = initial_ssthresh
    cwnd_history = [cwnd]
    ssthresh_history = [ssthresh]
    states = ['SS']
    for rtt in range(1, max_rounds):
        if loss_at and rtt in loss_at:
            ssthresh = max(cwnd // 2, 1)
            if loss_at[rtt] == 'timeout':
                cwnd = 1
                states.append('SS')
            else:
                cwnd = ssthresh + 3
                states.append('FR')
        else:
            if cwnd < ssthresh:
                cwnd *= 2
                states.append('SS')
            else:
                cwnd += 1
                states.append('CA')
        cwnd_history.append(cwnd)
        ssthresh_history.append(ssthresh)
    return cwnd_history, ssthresh_history, states

cwnd_h, ssthresh_h, states = simulate_tcp(
    initial_ssthresh=16, max_rounds=25,
    loss_at={8: '3dup', 17: 'timeout'}
)

fig, ax = plt.subplots(figsize=(12, 6))
colors = {'SS': '#2196F3', 'CA': '#4CAF50', 'FR': '#FF9800'}
state_colors = [colors.get(s, '#9E9E9E') for s in states]

for i in range(len(cwnd_h)-1):
    ax.plot([i, i+1], [cwnd_h[i], cwnd_h[i+1]],
            color=state_colors[i+1], linewidth=2.5)
    ax.scatter(i, cwnd_h[i], color=state_colors[i], s=80, zorder=5)
ax.scatter(len(cwnd_h)-1, cwnd_h[-1], color=state_colors[-1], s=80, zorder=5)

ax.step(range(len(ssthresh_h)), ssthresh_h, where='post',
        linestyle='--', color='red', alpha=0.7, linewidth=1.5)

```

```

ax.axvline(x=8, color='orange', linestyle=':', alpha=0.8, linewidth=1.5)
ax.axvline(x=17, color='purple', linestyle=':', alpha=0.8, linewidth=1.5)
ax.text(8.2, max(cwnd_h)*0.85, '3 Dup ACK\n→ Fast Recovery', fontsize=9, color='darkorange')
ax.text(17.2, max(cwnd_h)*0.85, 'Timeout\n→ SS restart', fontsize=9, color='purple')

ax.set_xlabel('Transmission Round (RTT)', fontsize=12)
ax.set_ylabel('Congestion Window (MSS)', fontsize=12)
ax.set_title('TCP Reno: cwnd Evolution\n(RTT=8: 3 Dup ACK → Fast Recovery | RTT=17: Timeout → Slow Start)')
ax.grid(True, alpha=0.3)

patches = [
    mpatches.Patch(color='#2196F3', label='Slow Start (SS): ACK +1MSS → RTT 2'),
    mpatches.Patch(color='#4CAF50', label='Congestion Avoidance (CA): RTT +1MSS'),
    mpatches.Patch(color='#FF9800', label='Fast Recovery (FR)'),
    mpatches.Patch(color='red', label='ssthresh'),
]
ax.legend(handles=patches, loc='upper left', fontsize=9)
plt.tight_layout()
plt.show()

```

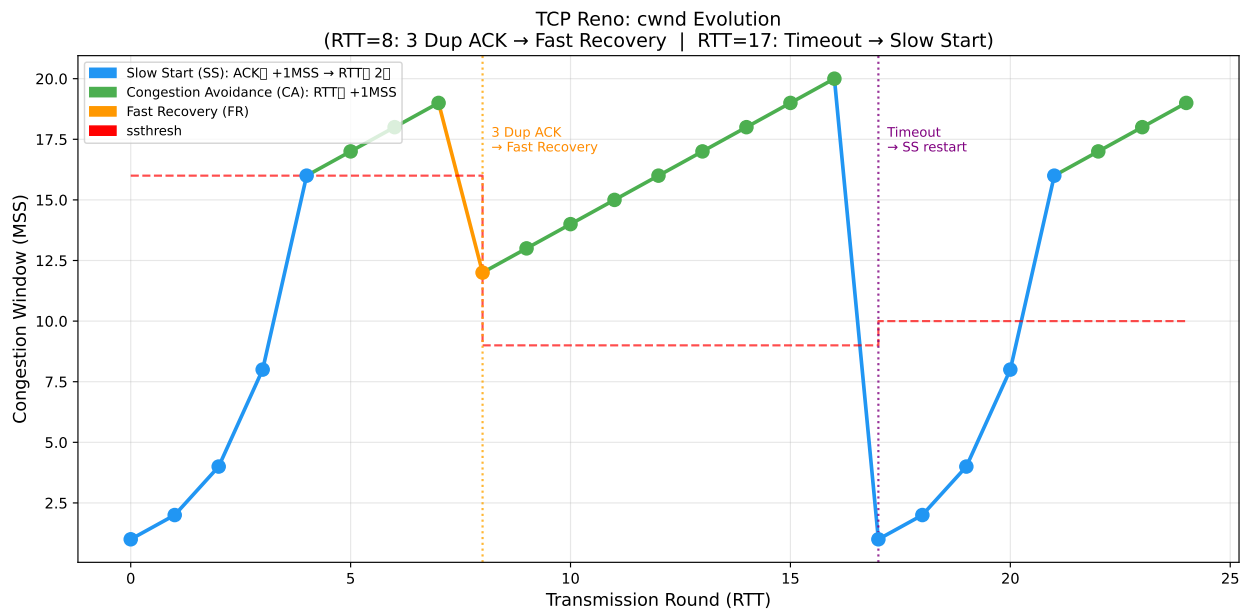


Figure 1: TCP Reno: cwnd (SS→CA→FR→SS)

4.2 Congestion Avoidance (CA)

$cwnd \geq ssthresh$, $cwnd$ (linear) .

:

- ACK : $cwnd += \frac{MSS^2}{cwnd}$
- : RTT $cwnd + 1 \text{ MSS}$ (Additive Increase)
- : $ssthresh = cwnd/2, cwnd \leftarrow cwnd/2$ (Multiplicative Decrease)

(linear) ? — $MSS^2/cwnd$

CA RTT +1 MSS . ACK :

$$\underbrace{\frac{cwnd}{MSS}}_{\text{RTT ACK}} \times x = \text{MSS} \Rightarrow x = \frac{MSS^2}{cwnd}$$

$MSS^2/cwnd$ “RTT +1 MSS” .

$$\frac{MSS^2}{cwnd} = MSS \times \underbrace{\frac{MSS}{cwnd}}_{cwnd - 1}$$

$cwnd - \text{ACK}$. : MSS = 1,460 bytes, $cwnd = 14,600$ bytes (10 segments) :

$$\frac{1460^2}{14600} = 146 \text{ bytes/ACK}$$

10 ACK $146 \times 10 = 1,460 = \text{MSS} \rightarrow \text{RTT} + 1 \text{ MSS}$.

SS CA :

	Slow Start	Congestion Avoidance
ACK 1	+1 MSS	$+MSS^2/cwnd$ ()
RTT 1	$+cwnd$ () (exponential)	+1 MSS () (linear)

CA , $ssthresh$ “ ” .

AIMD (Additive Increase, Multiplicative Decrease) .

AIMD: $\underbrace{cwnd += 1 \text{ MSS per RTT}}_{\text{Additive Increase}}, \underbrace{cwnd \leftarrow cwnd/2 \text{ on loss}}_{\text{Multiplicative Decrease}}$

💡 Tip

AIMD

, AIMD throughput **45° (equal share line)** .

- **AI** : (+1, +1) — (45°)
- **MD** : —

throughput () () .

4.3 Fast Retransmit & Fast Recovery (FR)

Fast Retransmit: 3 Duplicate ACK , . Dup ACK “ ” ,
timeout .

Fast Recovery (TCP Reno): Fast Retransmit Slow Start , ssthresh Congestion
Avoidance .

```
# Fast Recovery : 3 Dup ACKs
ssthresh = cwnd / 2
cwnd = ssthresh + 3 # +3: 3 Dup ACK
→ Fast Recovery
- Dup ACK : cwnd += 1 MSS
- ACK : cwnd = ssthresh → CA
- Timeout : cwnd = 1 MSS → SS
```

Note

+3 : 3 Dup ACK “3” . in-flight .

5 TCP Tahoe vs. TCP Reno

	TCP Tahoe	TCP Reno
Timeout	ssthresh=cwnd/2, cwnd=1, SS	ssthresh=cwnd/2,
3 Dup ACK	ssthresh=cwnd/2, cwnd=1, SS	cwnd=ssthresh+3, Fast Recovery
	3 Dup ACK = Timeout	3 Dup ACK = ()

Note

TCP Reno Tahoe
3 Dup ACK . cwnd 1 MSS . Reno Fast Recovery throughput
. Reno Tahoe throughput loss .

```

import matplotlib.pyplot as plt

rounds      = list(range(0, 20))
cwnd_tahoe  = [1, 2, 4, 8, 9, 10, 11, 12, 1, 2, 4, 6, 8, 9, 10, 11, 12, 13, 14, 15]
cwnd_reno   = [1, 2, 4, 8, 9, 10, 11, 12, 9, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]
sssthresh   = [8]*8 + [6]*12

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(rounds, cwnd_tahoe, 'b-o', markersize=6, linewidth=2, label='TCP Tahoe')
ax.plot(rounds, cwnd_reno, 'g-s', markersize=6, linewidth=2, label='TCP Reno')
ax.step(range(20), sssthresh, 'r--', where='post', linewidth=1.5, alpha=0.8, label='sssthresh')
ax.axvline(x=8, color='orange', linestyle=':', linewidth=2, label='3 Dup ACK @ round 8 (cwnd=12)')

# Reno
ax.fill_between(range(8, 14),
                cwnd_tahoe[8:14], cwnd_reno[8:14],
                alpha=0.15, color='green', label='Reno')

ax.annotate('Tahoe: cwnd→1\n(SS restart)',
            xy=(8, 1), xytext=(9.5, 2.5),
            arrowprops=dict(arrowstyle='->', color='blue'), fontsize=9, color='blue')
ax.annotate('Reno: cwnd→sssthresh+3=9\n(Fast Recovery)',
            xy=(8, 9), xytext=(9.5, 11),
            arrowprops=dict(arrowstyle='->', color='green'), fontsize=9, color='green')

ax.set_xlabel('Transmission Round', fontsize=12)
ax.set_ylabel('cwnd (MSS)', fontsize=12)
ax.set_title('TCP Tahoe vs. Reno: cwnd Evolution\n(3 Dup ACKs at Round 8, cwnd=12 → sssthresh=6)')
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
ax.set_ylim(0, 20)
plt.tight_layout()
plt.show()

```

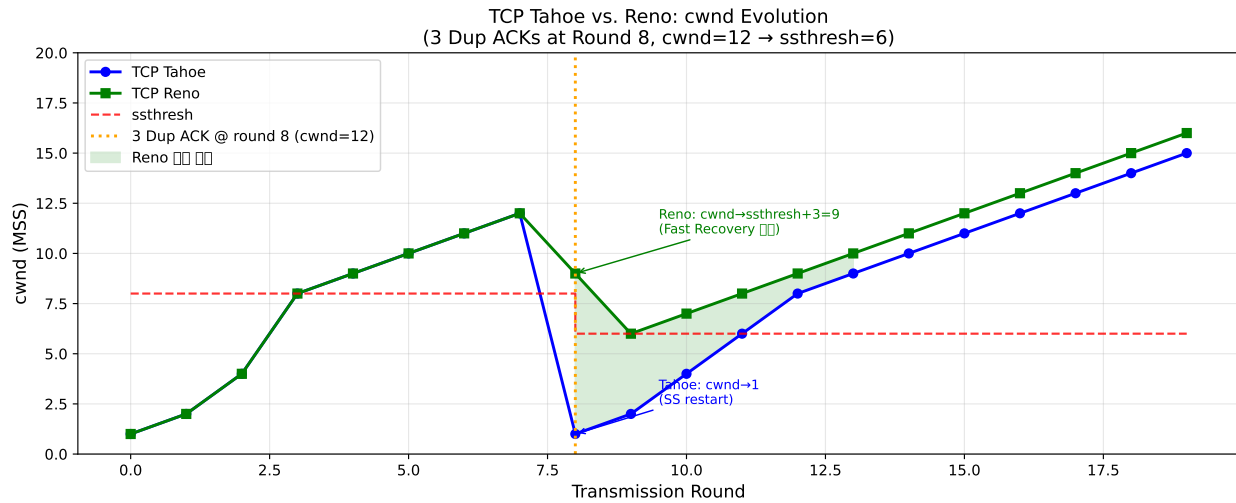


Figure 2: TCP Tahoe vs. Reno: 3 Dup ACK cwnd (ssthresh=8, loss at cwnd=12)

6 TCP FSM

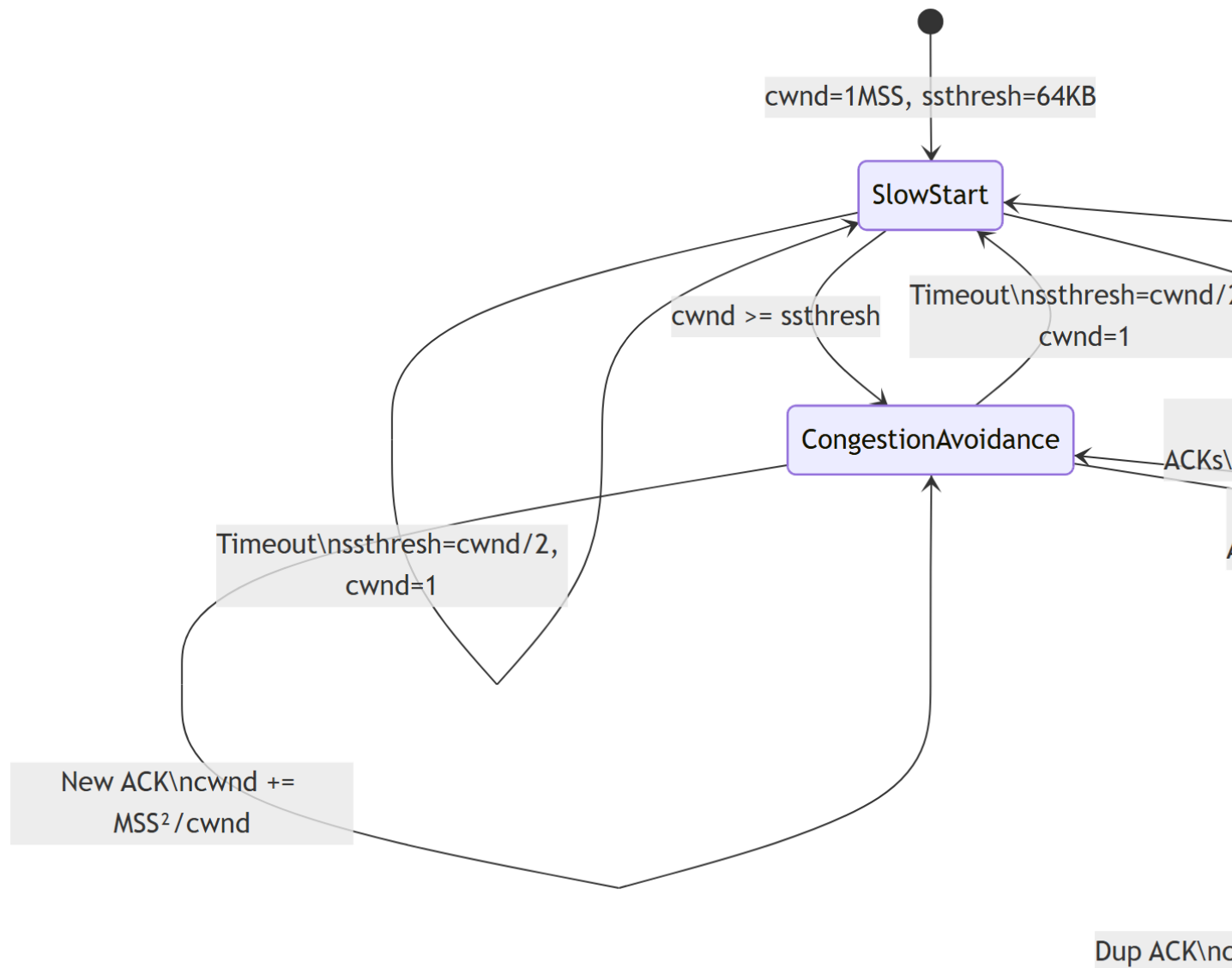
TCP Reno Finite State Machine .

```
stateDiagram-v2
    [*] --> SlowStart : cwnd=1MSS, ssthresh=64KB

    SlowStart --> SlowStart : New ACK\ncwnd += MSS
    SlowStart --> CongestionAvoidance : cwnd >= ssthresh
    SlowStart --> SlowStart : Timeout\nssthresh=cwnd/2, cwnd=1
    SlowStart --> FastRecovery : 3 Dup ACKs\nssthresh=cwnd/2\ncwnd=ssthresh+3

    CongestionAvoidance --> CongestionAvoidance : New ACK\ncwnd += MSS2/cwnd
    CongestionAvoidance --> SlowStart : Timeout\nssthresh=cwnd/2, cwnd=1
    CongestionAvoidance --> FastRecovery : 3 Dup ACKs\nssthresh=cwnd/2\ncwnd=ssthresh+3

    FastRecovery --> FastRecovery : Dup ACK\ncwnd += MSS
    FastRecovery --> CongestionAvoidance : New ACK\ncwnd=ssthresh
    FastRecovery --> SlowStart : Timeout\nssthresh=cwnd/2, cwnd=1
```



7 TCP Throughput

7.1 (AIMD)

AIMD $cwnd$ $W/2$ W . $cwnd = W$:

$$\overline{\text{throughput}} = \frac{3}{4} \cdot \frac{W}{RTT} \quad [\text{bytes/sec}]$$

$$: \quad = \frac{W/2 + W}{2} = \frac{3W}{4}$$

7.2 Mathis Formula ()

L [Mathis 1997]:

$$\overline{\text{throughput}} = \frac{1.22 \cdot MSS}{RTT \cdot \sqrt{L}}$$

💡 Tip

()
AIMD :

- Additive Increase: $W/2 \rightarrow W$ $W/2$ RTT
- : $\sum_{k=W/2}^W k \cdot MSS \approx \frac{3W^2}{8} \cdot MSS$
- $1 \rightarrow L = \frac{8}{3W^2}$
- $W = \sqrt{\frac{8}{3L}}$ $\text{throughput} = \frac{1.22 \cdot MSS}{RTT \sqrt{L}}$

7.3 Long Fat Pipe

(: 10 Gbps, RTT=100ms) :

$$L \leq 2 \times 10^{-10}$$

. CUBIC, BBR TCP .

```
import numpy as np
import matplotlib.pyplot as plt

MSS = 1500
RTT = 0.1
L_range = np.logspace(-10, -2, 500)
throughput_Mbps = (1.22 * MSS) / (RTT * np.sqrt(L_range)) / 1e6

fig, ax = plt.subplots(figsize=(10, 5))
ax.loglog(L_range, throughput_Mbps, 'b-', linewidth=2.5, label='Mathis Formula')

thresholds = {
    10000: ('10 Gbps', 'red'),
    1000: ('1 Gbps', 'orange'),
```

```

100: ('100 Mbps','green'),
10: ('10 Mbps', 'steelblue'),
}
for tp, (label, color) in thresholds.items():
    L_needed = (1.22 * MSS / (RTT * tp * 1e6))**2
    ax.axhline(y=tp, color=color, linestyle='--', alpha=0.5, linewidth=1)
    ax.axvline(x=L_needed, color=color, linestyle=':', alpha=0.5, linewidth=1)
    ax.text(L_needed * 1.5, tp * 1.4,
            f'{label}\nL={L_needed:.1e}', fontsize=8, color=color)

ax.set_xlabel('Packet Loss Rate (L)', fontsize=12)
ax.set_ylabel('Throughput (Mbps)', fontsize=12)
ax.set_title('TCP Throughput vs. Loss Rate\n(Mathis Formula, MSS=1500B, RTT=100ms)', fontsize=12)
ax.grid(True, which='both', alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()

```

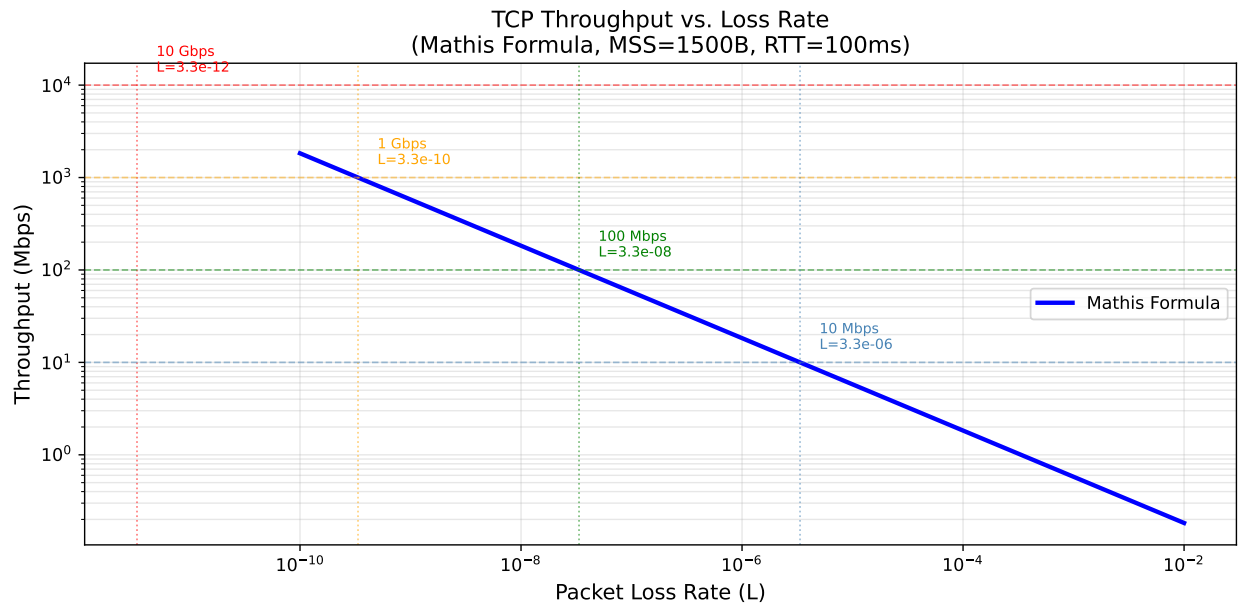


Figure 3: Mathis Formula: TCP Throughput (MSS=1500B, RTT=100ms)

8 TCP (Fairness)

8.1 AIMD

K TCP R , :

$$= \frac{R}{K}$$

:

- **AI** : (+1,+1) — (45°)
- **MD** : (0.5 · x₁, 0.5 · x₂) —

throughput () () .

```
import numpy as np
import matplotlib.pyplot as plt

R = 10
fig, ax = plt.subplots(figsize=(8, 8))

ax.fill_between([0, R], [R, 0], [R, R], alpha=0.05, color='red')
ax.fill_between([0, R], [0, 0], [R, 0], alpha=0.05, color='green')
ax.plot([0, R], [0, R], 'k--', linewidth=1.5, label='Equal share (45°)')
ax.plot([0, R], [R, 0], 'r-', linewidth=2, alpha=0.7, label='Capacity limit')

np.random.seed(42)
x1, x2 = 1.0, 7.0
trajectory = [(x1, x2)]
for _ in range(20):
    x1 += 1; x2 += 1
    if x1 + x2 > R:
        x1 *= 0.5; x2 *= 0.5
    trajectory.append((x1, x2))

xs, ys = zip(*trajectory)
ax.plot(xs, ys, 'b-o', markersize=5, linewidth=1.5, alpha=0.8, label='AIMD trajectory')
ax.scatter(xs[0], ys[0], s=150, c='blue', marker='^', zorder=6, label='Start')
ax.scatter(xs[-1], ys[-1], s=150, c='red', marker='*', zorder=6, label='Converged')

mid_idx = 5
ax.annotate('', xy=(xs[mid_idx]+1, ys[mid_idx]+1),
            xytext=(xs[mid_idx], ys[mid_idx]),
            arrowprops=dict(arrowstyle='->', color='green', lw=2))
ax.text(xs[mid_idx]+0.3, ys[mid_idx]+0.2, 'AI (+1,+1)', fontsize=9, color='green')

ax.annotate('', xy=(xs[mid_idx+1]*0.5, ys[mid_idx+1]*0.5),
            xytext=(xs[mid_idx+1], ys[mid_idx+1]),
            arrowprops=dict(arrowstyle='->', color='red', lw=2))
ax.text(xs[mid_idx+1]*0.5+0.2, ys[mid_idx+1]*0.5+0.3, 'MD (×0.5)', fontsize=9, color='red')

ax.set_xlim(0, R+1); ax.set_ylim(0, R+1)
ax.set_xlabel('Throughput of Connection 1', fontsize=12)
```



```

ax.set_ylabel('Throughput of Connection 2', fontsize=12)
ax.set_title('AIMD Fairness Convergence\n(Two TCP Connections, Link Capacity R=10)', fontsize=12)
ax.legend(fontsize=10, loc='upper right')
ax.grid(True, alpha=0.3)
ax.set_aspect('equal')
plt.tight_layout()
plt.show()

```

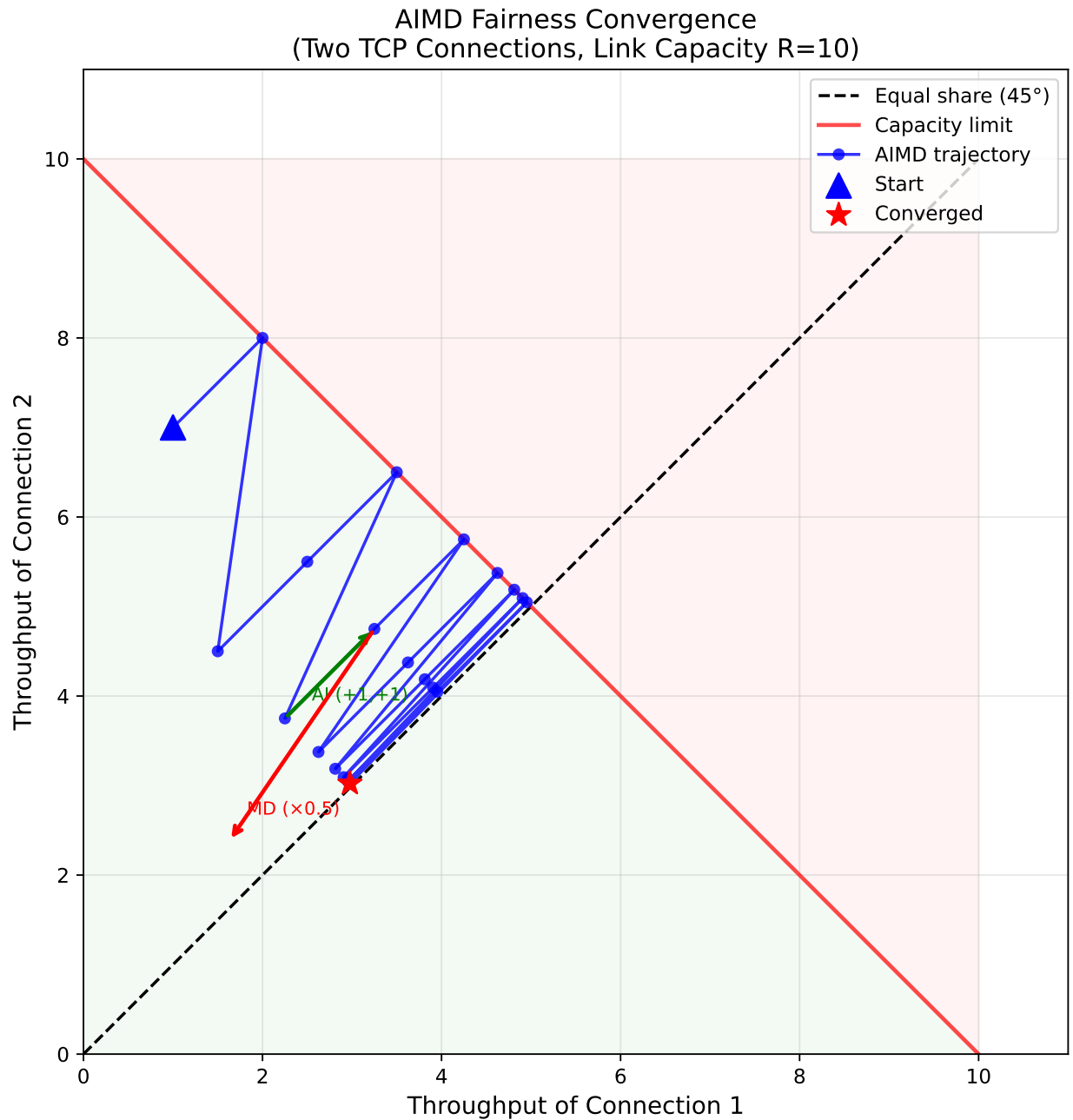


Figure 4: AIMD Fairness: TCP Throughput

8.2

- **RTT** : RTT $\xrightarrow{cwnd}$ RTT throughput
- **TCP** : N N
- **UDP** : UDP TCP

9 Explicit Congestion Notification (ECN)

TCP (loss) , ECN (network-assisted) .

9.1

[] $\rightarrow$ []
IP datagram: ECN=10 (capable)
 $\downarrow$ ()
IP datagram: ECN=11 (CE: Congestion Experienced)
[] $\leftarrow$ []
TCP ACK: ECE=1 (Echo CE)
[]: cwnd , CWR=1

ECN	
00	ECN
10 or 01	ECN (ECT: ECN-Capable Transport)
11	(CE: Congestion Experienced)

i Note

ECN : , . (DCTCP) .

10

10.1

$rate \approx cwnd / RTT$

throughput	$\bar{T} = \frac{3}{4} \cdot \frac{W}{RTT}$
Mathis throughput	$\bar{T} = \frac{1.22 \cdot MSS}{RTT \cdot \sqrt{L}}$
CA ACK	$\Delta cwnd = MSS^2 / cwnd$ $= R/K$

10.2 cwnd

		cwnd	
Slow Start	New ACK	cwnd += 1 MSS	ACK → RTT 2
Slow Start	Timeout	ssthresh=cwnd/2, cwnd=1	
Slow Start	3 Dup ACK	ssthresh=cwnd/2, cwnd=ssthresh+3 → FR	Reno
Congestion Avoidance	New ACK	cwnd += MSS ² /cwnd	RTT +1 MSS
Congestion Avoidance	Timeout	ssthresh=cwnd/2, cwnd=1	
Congestion Avoidance	3 Dup ACK	ssthresh=cwnd/2, cwnd=ssthresh+3 → FR	
Fast Recovery	Dup ACK	cwnd += 1 MSS	
Fast Recovery	New ACK	cwnd=ssthresh → CA	
Fast Recovery	Timeout	ssthresh=cwnd/2, cwnd=1 → SS	

10.3

: ssthresh = 8 MSS, cwnd = 1 MSS TCP Reno
3 Dup ACK, cwnd = 12 MSS.

1. ssthresh ?
2. Fast Recovery cwnd ?
3. ACK cwnd ?

cwnd ():

Round	cwnd	
1 SS	1	
2 SS	2	ACK +1 → RTT 2
3 SS	4	

Round	cwnd		
4	SS→CA	8 = ssthresh → CA	cwnd ssthresh
5	CA	9	RTT +1
6	CA	10	
7	CA	11	
8	CA	12	→ 3 Dup ACK

:

Q1. ssthresh:

$$ssthresh = \frac{cwnd}{2} = \frac{12}{2} = \boxed{6 \text{ MSS}}$$

Q2. Fast Recovery cwnd (TCP Reno):

$$cwnd = ssthresh + 3 = 6 + 3 = \boxed{9 \text{ MSS}}$$

Q3. Missing segment ACK :

$$cwnd = ssthresh = \boxed{6 \text{ MSS}}, \rightarrow \text{Congestion Avoidance}$$

```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

rounds = list(range(0, 16))
cwnd = [1, 2, 4, 8, 9, 10, 11, 12, 9, 6, 7, 8, 9, 10, 11, 12]
sst = [8]*8 + [6]*8
states = ['SS', 'SS', 'SS', 'SS', 'CA', 'CA', 'CA', 'CA', 'FR', 'CA', 'CA', 'CA', 'CA', 'CA', 'CA', 'CA']

colors = {'SS': '#2196F3', 'CA': '#4CAF50', 'FR': '#FF9800'}
fig, ax = plt.subplots(figsize=(12, 5))

for i in range(len(cwnd)-1):
    ax.plot([rounds[i], rounds[i+1]], [cwnd[i], cwnd[i+1]],
            color=colors[states[i+1]], linewidth=2.5)
    ax.scatter(rounds[i], cwnd[i], color=colors[states[i]], s=70, zorder=5)
ax.scatter(rounds[-1], cwnd[-1], color=colors[states[-1]], s=70, zorder=5)

ax.step(rounds, sst, 'r--', where='post', linewidth=1.5, alpha=0.8)

ax.axvline(x=7, color='orange', linestyle=':', linewidth=2)
ax.annotate('3 Dup ACK\ncwnd=12\n→ ssthresh=6, cwnd=9 (FR)',
            xy=(7, 12), xytext=(8.5, 13),
            arrowprops=dict(arrowstyle='->', color='darkorange'),
            fontsize=9, color='darkorange',
```

```

        bbox=dict(boxstyle='round,pad=0.3', facecolor='lightyellow', alpha=0.8))

ax.annotate('New ACK\ncwnd = ssthresh = 6\n→ CA ',
            xy=(9, 6), xytext=(10.5, 3.5),
            arrowprops=dict(arrowstyle='->', color='green'),
            fontsize=9, color='darkgreen')

ax.set_xlabel('Transmission Round', fontsize=12)
ax.set_ylabel('cwnd (MSS)', fontsize=12)
ax.set_title(' : TCP Reno cwnd \n(ssthresh=8, 3 Dup ACK at cwnd=12 → ssthresh=6, FR CA
ax.grid(True, alpha=0.3)
ax.set_ylim(0, 16)

patches = [
    mpatches.Patch(color='#2196F3', label='Slow Start (SS)'),
    mpatches.Patch(color='#4CAF50', label='Congestion Avoidance (CA)'),
    mpatches.Patch(color='#FF9800', label='Fast Recovery (FR)'),
    mpatches.Patch(color='red', label='ssthresh'),
]
ax.legend(handles=patches, loc='upper left', fontsize=9)
plt.tight_layout()
plt.show()

```

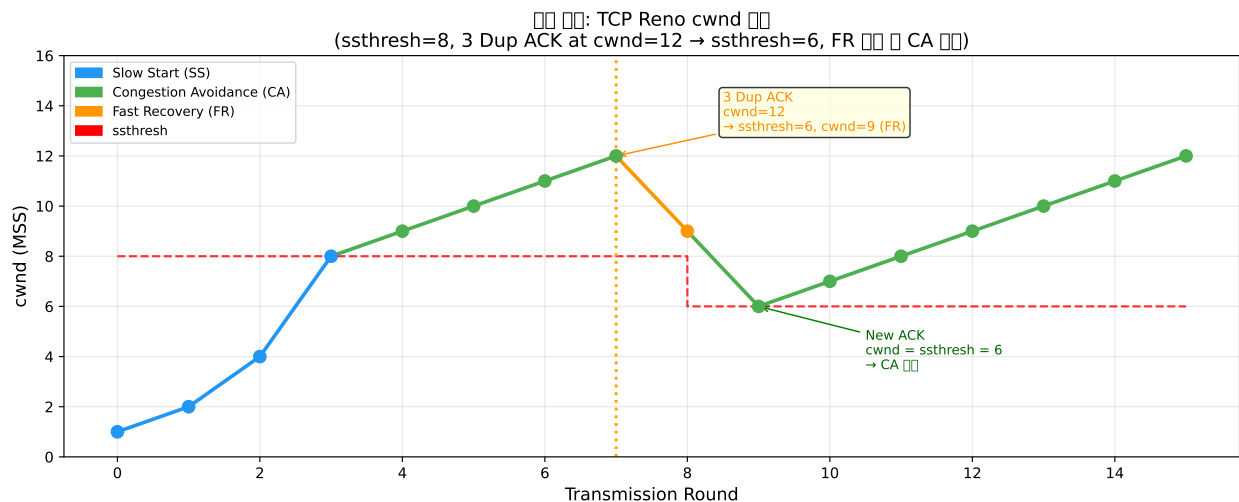


Figure 5: : ssthresh=8, 3 Dup ACK at cwnd=12

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- Chapter 3.6: Principles of Congestion Control
 - Chapter 3.7: TCP Congestion Control
- Jacobson, V. (1988). Congestion Avoidance and Control. *ACM SIGCOMM*.
- Mathis, M. et al. (1997). The Macroscopic Behavior of the TCP Congestion Avoidance Algorithm. *ACM SIGCOMM Computer Communication Review*.
- RFC 5681 — TCP Congestion Control.
- RFC 3168 — The Addition of Explicit Congestion Notification (ECN) to IP.